

MISSION 5 – MULTI-DRONE OPERATION

Objective

To demonstrate simultaneous multi-drone operation using two SITL instances, monitored and controlled via both MAVProxy and QGroundControl, with independent waypoint missions assigned to each vehicle.

Setup & Configuration

Two SITL instances were launched from a single terminal using the `--count 2` flag, which automatically spawned both vehicles with auto-assigned system IDs:

```
sim_vehicle.py --count 2 --auto-sysid
```

- **Vehicle 1:** SYSID = 1 — assigned 3 waypoints inside Tech Green
- **Vehicle 2:** SYSID = 2 — assigned 3 waypoints along the Tech Green boundary

Both vehicles shared **UDP port 14550**, as QGroundControl only supports a single UDP connection and cannot run multiple instances. QGroundControl automatically detected both vehicles on the same port and displayed them simultaneously on the map, allowing switching between them directly within the GCS interface.

In MAVProxy, switching between vehicles was done using:

```
vehicle 1    ← directs commands to Vehicle 1 (SYSID=1)
vehicle 2    ← directs commands to Vehicle 2 (SYSID=2)
```

See video at T = 0:39 for MAVProxy vehicle switching while both drones operate in parallel.

Observations

- Both vehicles were successfully spawned from a single terminal using `--count 2` with auto-assigned SYSIDs, confirming that multi-drone SITL setup does not require separate terminal instances.

- Both vehicles shared UDP port 14550 — QGroundControl automatically detected both SYSIDs on the same port and displayed them simultaneously on the map with independent telemetry streams for each.
- Switching between vehicles in QGroundControl was seamless — both drones were visible at all times and the active vehicle could be selected directly within the GCS interface.
- MAVProxy switching between vehicles using `vehicle 1` and `vehicle 2` commands correctly directed commands to the intended SYSID without affecting the other vehicle.
- Minor launch delay occurred due to manual switching during initialization.
- Both missions completed successfully — Vehicle 1 completed its interior Tech Green waypoints and Vehicle 2 completed its boundary waypoints without interference.

Conclusion

Simultaneous multi-drone operation is feasible when the following conditions are met:

- **Single terminal launch with --count 2** — Both SITL instances were spawned together with automatically assigned SYSIDs, simplifying the setup process.
- **Shared UDP Port** — Both vehicles operated over port 14550; QGroundControl's automatic multi-vehicle detection eliminated the need for separate ports or multiple GCS instances.
- **Proper GCS Switching** — QGroundControl natively handled both vehicles on the same connection, while MAVProxy's `vehicle 1` / `vehicle 2` commands provided precise command routing to each SYSID independently. MAVProxy switching is demonstrated in the video at T = 0:39 .